

Table 2: Performance comparison between direct answering of LLMs and long-CoT LRMs. *Token cost represents the weighted sum of context and inference tokens with a 0.25:1 ratio.

Models		MiniARC	COGS	RAVEN	Average	
					Accuracy (%)	Token Cost*
LLM (direct)	Qwen2.5-32B	24.16	21.80	12.07	19.34	207.44
	Qwen2.5-72B	23.49	20.40	23.67	22.52	198.54
	Gemini-1.5-pro	24.83	36.00	<u>24.31</u>	28.38	198.95
	Llama-3.1-405B	26.71	<u>24.40</u>	<u>25.34</u>	25.48	201.61
	Deepseek-V3	<u>27.52</u>	<u>30.80</u>	21.05	<u>26.46</u>	189.71
LRM (long-CoT)	QwQ-32B	18.70	13.00	8.82	13.51	1736.91
	o1-mini	30.20	10.60	15.25	18.68	3072.02
	Deepseek-R1	<u>28.86</u>	24.00	27.56	<u>26.81</u>	2432.36

on MiniSCAN—than the converse scenario. This disparity underscores the dominance of implicit reasoning, which frequently compensates for flawed explicit pattern inference to achieve a correct final answer.

Findings: Implicit reasoning significantly outweighs explicit reasoning in contributing to CoT success, validating the asymmetric hybrid mechanism proposed in Hypothesis 4.

Summary: Our analysis suggests a compelling explanation for CoT’s underperformance in this domain: it operates as a dual-process mechanism, combining both explicit and implicit reasoning (Hypothesis 4), but this mechanism is fundamentally compromised. First, its explicit reasoning pathway appears unreliable, primarily because LLMs struggle to correctly infer underlying patterns from demonstrations (Hypothesis 2). Second, the very structure of CoT—the insertion of a rationale—increases contextual distance, which degrades the performance of its implicit reasoning pathway (Hypothesis 1), a pathway that otherwise appears robust. With its explicit component proving ineffective and its implicit component hampered, CoT consistently underperforms direct answering, a method that relies solely on an uncompromised implicit process.

6 The case of specialized reasoning models

A crucial test of our hypothesis involves Large Reasoning Models (LRMs), which are explicitly designed to excel at complex, multi-step reasoning. In pattern-based ICL, a key distinction between these LRMs and traditional LLMs lies in the former’s ability to iteratively propose and refine hypothesized patterns, thereby enhancing explicit pattern inference capabilities. However, this extended reasoning process significantly increases contextual distance, creating a direct trade-off with the implicit reasoning pathway.

To evaluate this trade-off, we conducted experiments using three LRMs across three benchmarks, each selected from a distinct task modality. As shown in Table 2, the results are striking: top-performing LLMs using direct answering achieve **comparable or superior performance** to the specialized LRMs. This outcome is particularly notable given that the LRMs consume, on average, **12× more total tokens and 40× more inference tokens**.

These findings suggest that even when the explicit reasoning component is significantly enhanced, its benefits are insufficient to overcome the structural disadvantages imposed by the CoT framework in these tasks. The performance of the implicit pathway, hampered by increased contextual distance, remains a critical bottleneck. This underscores the severity of the "CoT curse" in this domain and highlights the need for strategies that can integrate verbalized and latent reasoning more efficiently.

7 Related Work

Inductive Reasoning Our pattern-based in-context learning setup aligns with inductive reasoning: models must infer an implicit pattern from demonstration pairs and apply it to a test instance. Inductive reasoning capabilities in large language models have been extensively studied, with a focus on the design of reasoning

frameworks and training or prompting strategies that elicit such behavior (Wang et al., 2024; Qiu et al., 2024; Liu et al., 2024; Zheng et al., 2025a; Li et al., 2025). These capabilities play a critical role across diverse application domains, including financial modeling (Goel et al., 2025; Stempień & Ślepaczuk, 2025) and scientific discovery (Shojaee et al., 2025; Zheng et al., 2025b;c).

Effectiveness of Chain-of-Thought Prompting Chain-of-Thought prompting (Wei et al., 2022) is widely recognized for improving reasoning performance. At the same time, evidence suggests that CoT offers limited gains for tasks that are less reasoning-intensive, such as factuality calibration (Zong et al., 2025b;a; Liu et al., 2025) and semantic classification (Sprague et al., 2024). Recent work has systematically examined how CoT length affects performance (Jin et al., 2024; Wu et al., 2025; Nayab et al., 2025). In contrast, our work provides the first evidence, to our knowledge, that CoT exhibits substantial limitations on a reasoning-intensive task setting where success depends on inferring and generalizing latent patterns from demonstrations.

8 Conclusion

In this work, we identify and rigorously analyze a fundamental paradox in Chain-of-Thought prompting: despite its success in various reasoning tasks, CoT consistently underperforms direct answering in pattern-based in-context learning—a fundamental task that harnesses the inductive and abstract reasoning capabilities of LLMs. Our investigation reveals that this failure stems from a flawed **explicit-implicit hybrid mechanism** at the core of CoT’s mechanism. We find that the explicit reasoning pathway, responsible for articulating rationales, is unreliable, as large language models struggle to correctly infer underlying patterns from demonstrations. Simultaneously, the very structure of CoT—the insertion of these rationales—increases contextual distance, degrading the performance of the otherwise robust implicit reasoning pathway that powers direct answering. With its explicit component proving ineffective and its implicit component hampered, CoT is fundamentally compromised in these tasks.

Our analysis of specialized Large Reasoning Models further validates this conclusion, demonstrating that even architectures designed for superior explicit reasoning fail to overcome these inherent limitations, performing on par with or worse than direct answering despite incurring substantial computational overhead. These findings challenge the presumed universal efficacy of CoT, highlighting the critical need for a more nuanced understanding that balances explicit and implicit processes. We advocate for the development of adaptive reasoning strategies that can harness the strengths of both modes, paving the way for more robust and efficient large language models.

Acknowledgement

We thank all the anonymous reviewers and editor for their valuable comments. The authors of this paper were supported by the ITSP Platform Research Project (ITS/189/23FP) from ITC of Hong Kong, SAR, China, and the AoE (AoE/E-601/24-N), the RIF (R6021-20) and the GRF (16205322) from RGC of Hong Kong, SAR, China. We also thank the support from NVIDIA AI Technology Center (NVAITC).

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